



Cambridge IGCSE™

PHYSICS

0625/43

Paper 4 Extended Theory

May/June 2023

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Acronyms and shorthand in the mark scheme

Acronym / shorthand	Explanation
A mark	Final answer mark which is awarded for fully correct final answers including the unit.
C mark	Compensatory mark which may be scored when the final answer (A) mark for a question has not been awarded.
B mark	Independent mark which does not depend on any other mark.
M mark	Method mark which must be scored before any subsequent final answer (A) mark can be scored.
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word / phrase / unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.
owtte	Or words to that effect.
ignore	Indicates either an incorrect or irrelevant point which may be disregarded, i.e., <u>not</u> treated as contradictory.
insufficient	An answer not worthy of credit <u>on its own</u> .
CON	An incorrect point which contradicts any correct point and means the mark cannot be scored.
ecf [question part]	Indicates that a candidate using an erroneous value from the stated question part must be given credit here if the erroneous value is used correctly here.
cao	Correct answer only.
ORA	Or reverse argument.

Question	Answer	Marks
1(a)	1200 N AND $g = W / m$ OR $(W =) mg$ OR $(W =) 120 \times 9.8$	B1
1(b)	$F = ma$ OR $(a =) F / m$	B1
	$(a =) 54 / 120$	B1
1(c)	$(v =) 3.6 \text{ m/s}$	A2
	$a = (\Delta)v / t$ OR $(\Delta v =) at$ OR $(\Delta v =) 0.45 \times 8(.0)$	C1
1(d)	$(d =) 14 \text{ m}$	A2
	<u>average</u> speed = (total) distance travelled / (total) time taken OR 1.8×8 OR $\left(\frac{3.6}{2}\right) \times 8$	C1

Question	Answer	Marks
2(a)(i)	(pressure is) force per unit area	B1
2(a)(ii)	(variation with depth) increases (as depth increases)	B1
	(variation with density) increases (as density increases)	B1
2(b)	any two from: • geothermal • nuclear • tidal	B2
2(c)(i)	(statement) non-renewable / not renewable / no	B1
	(explanation) (nuclear) fuel is used up	B1
2(c)(ii)	(statement) renewable / yes	B1
	(explanation) waves will always continue OR produced by wind which will always continue OR nothing used up	B1

Question	Answer	Marks
3(a)(i)	(greatest) gas	B1
	(least) solid	B1
3(a)(ii)	any three from: • (solids) particles vibrate • (gases) particles move freely • (solids) particles in fixed / close positions • (gases) particles randomly arranged (in container) / wide separation • (gas) particles move quickly	B3
3(b)(i)	(specific heat capacity is the) energy required to raise 1 kg / unit mass by 1 °C / 1 K / 1 kelvin / unit temperature	A2
	energy required per unit mass / 1 kg OR energy required per unit temperature / 1 °C increase / 1 K increase / 1 kelvin increase	C1
3(b)(ii)	time	B1
	mass	B1
	initial temperature AND final temperature	B1

Question	Answer	Marks
4(a)(i)	two correct rays extended back (towards the intersection)	M1
	extended rays intersect	A1
	image drawn from intersection to principal axis AND base of image lies in correct range	B1
4(a)(ii)	circles around: • enlarged • virtual • upright	B3
4(b)(i)	long-sightedness / long sight / far-sighted	B1
4(b)(ii)	converging lens reduces focal length of eye OR converging lens brings focal point forward OR without lens, rays converge behind back of eye	B1
	(so that) rays converge / focus on back of eye / retina	B1

Question	Answer	Marks
5(a)(i)	infrared	B1
5(a)(ii)	glass is transparent to visible light and (some) IR	B1
	(visible light and some IR) can carry high rates of data / information	B1
5(b)(i)	$(n =) 1.4$	A2
	$(n =) 1 / \sin c$ OR $(n =) 1 / \sin c$ OR $(n =) 1 / \sin 45$	C1
5(b)(ii)	diagram shows that total internal reflection is taking place inside optical fibre	B1
	one or more <u>correct</u> reflections with $i > 45^\circ$	B1

Question	Answer	Marks
6(a)	$(I =) 9.8 \text{ A}$	A3
	$P = I^2 R \text{ OR } I^2 = \frac{P}{R} \text{ OR } I^2 = \frac{2500}{26}$	C1
	$I^2 = 2500 / 26 \text{ OR } (I =) \sqrt{(P / R)} \text{ OR } \sqrt{\frac{2500}{26}}$	C1
6(b)	$(R =) 53 \Omega$	A3
	$R \propto l \text{ OR } 1.8 / 1.2 \text{ OR } 1.5 \text{ seen as multiplier}$	C1
	$R \propto 1 / A \text{ OR } 7.9 (\times 10^{-7}) / 5.8 (\times 10^{-7}) \text{ OR } 1.36(2) \text{ seen as multiplier}$	C1
6(c)	$(\text{cost} =) \$36$	A2
	$E = Pt \text{ AND } (\text{cost} =) E \times 0.3(0) \text{ OR } (\text{cost} =) 25 \times 10^N \times 2 \times 24 \times 0.3(0) \text{ OR } 3.6 \times 10^N \text{ dollars}$	C1

Question	Answer	Marks
7(a)	$(I_p =) 0.14 \text{ A}$	A3
	$V_p I_p = V_s I_s$ OR $(I_p =) V_s I_s / V_p$ OR $(I_p =) 12 \times 2.5 / 220$	C1
	$(I_p =) 12 \times 2.5 / 220$	C1
7(b)	(turns ratio =) 18 (: 1)	A2
	$N_p / N_s = V_p / V_s$ OR $(N_p / N_s =) V_p / V_s$ OR (turns ratio =) 220 / 12	C1

Question	Answer	Marks
8(a)(i)	(neutron) becomes proton and electron OR (neutron) becomes proton and β particle	B1
8(a)(ii)	charge on neutron = 0 OR total charge on products = 0	B1
	charge on proton = +1 AND charge on electron = –1	B1
8(b)	$({}_{86}^{212}\text{Rn} \rightarrow) {}_{84}^{208}\text{Po} + {}_2^4\alpha$	A3
	any two from: - proton numbers balance - nucleon numbers balance ${}_2^4\alpha$ OR ${}_2^4\text{He}$ seen	C2

Question	Answer	Marks
9(a)	both positions correctly marked S ₁ and J ₁ on the diagram	A3
	Saturn moved $360 \times 5 / 30 (= 60^\circ)$ OR S ₁ in correct position	C1
	Jupiter moved $360 \times 5 / 12 (= 150^\circ)$ OR J ₁ in correct position	C1
9(b)(i)	S ₂ at 240° AND J ₂ at $(600^\circ - 360^\circ =) 240^\circ$	B1
9(b)(ii)	(Saturn and Jupiter) are aligned / Jupiter exactly in front of Saturn / there is a conjunction owtte	B1
9(c)(i)	(Jupiter) gaseous AND large AND (Earth) rocky AND small	B1
9(c)(ii)	any three from: (density) - Jupiter has a low density because it is composed of gas / Earth has a high density because it is a solid (gravitational field strength) - Jupiter (has a large GFS so it) has a large mass / Earth (has a small GFS so it) has a small mass - Jupiter's mass is larger than the Earth's mass because the volume of Jupiter is larger even though the density of Jupiter is smaller	B3
9(d)	(mass =) 1.8×10^{27} kg	A3
	$\rho = m / V$ OR $(m =) \rho V$ OR $(m =) 1300 \times 1.4 \times 10^{15} \times 10^9$	C1
	$(m =) 1300 \times 1.4 \times 10^{15} \times 10^9$ (kg) OR $1.8(2) \times 10^N$	C1

Question	Answer	Marks
10(a)	$v = s / t$ OR $(s =) vt$	B1
	$(c =) 3 \times 10^8 \text{ (m / s)}$	B1
	$(1 \text{ year} =) 365 \times 24 \times 3600 \text{ (s)}$ OR $3.2 \times 10^7 \text{ (s)}$ OR 32×10^6 OR 8760×3600	B1
	$(s =) \text{ candidate's speed of light} \times \text{candidate's time (m)}$	B1
10(b)(i)	change of wavelength (of galaxy's starlight) OR redshift	B1
10(b)(ii)	$H_0 = v / d$	B1
10(b)(iii)	brightness of a supernova	B1